

Question 1

1. Use the Waterfall model, but include prototyping to validate critical requirements early.

Key Activities:

- Specification: Clearly define all requirements in detail. Use prototypes to check usability and safety before development starts.
 - Design: Focus on creating a strong system architecture that ensures reliability, safety, and fault tolerance.
 - Validation: Perform thorough testing at all levels (unit, integration, system, and acceptance). Use formal verification if needed.
 - Evolution: Follow a strict change management process. Analyze the impact of any change, perform regression testing, and re-certify the system if required.
2. Requirement Validation:
 - a. Consistency
 - b. Verifiability
 - c. Realism.

Question 02:

1. Agile is suitable because requirements are uncertain and change frequently across institutions.
2. User stories may be too informal to handle conflicting or complex academic requirements.
3. Pair programming reduces defects through continuous peer review.
4. Time zone differences lead to poor communication and inconsistent Scrum meetings.
5. Frequent feedback can change priorities, requiring modification of existing features.

Question 03:

delta 1	delta 2	delta 3				
6	3	1				
7	5	2				
7	4	2		E1	E2	E3
2	0	0	R1	97	98	124
7	3	1	R2	104	106	116
7	2	2	R3	108	107	113

estimation sheet						
Module	Kamran	Nida	Bilal	Best	worst	average
AI Damage Assessment	14	15	15	15	14	14.5
Rescue Team Tracking	18	18	20	20	18	19
Emergency App	22	22	24	24	22	23
Command Dashboard	18	18	18	18	18	18
Resource Optimization	14	14	15	15	14	14.5
Alert System	22	20	21	22	20	21

